Insurance Fraud Detection using Active Learning

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Initial Objective + Evolution of Objectives

Core Objectives:

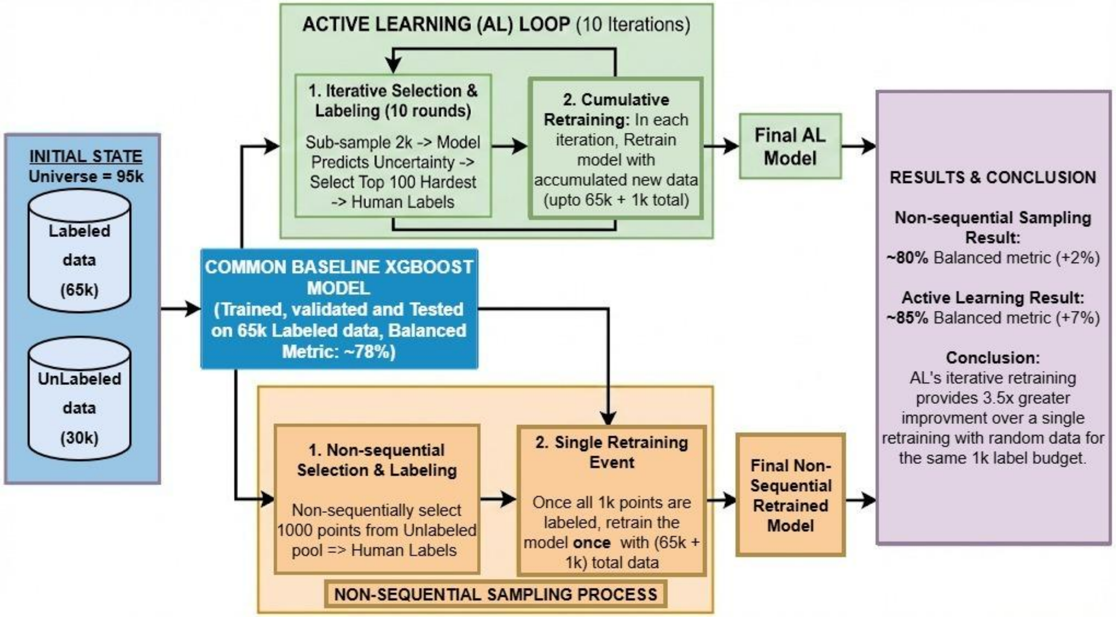
- Reduce financial loss in insurance streams for on ground use-cases.

Additional Objectives:

- Maximize ROI on labeling, to best spend a budget of new human labels to improve models to check if “what” we label matters equally as “how many”.
- To capture data distribution drift by allowing model to "learn as it goes" & refine its uncertainty with every batch.
- Accuracy based implementation for analysis.

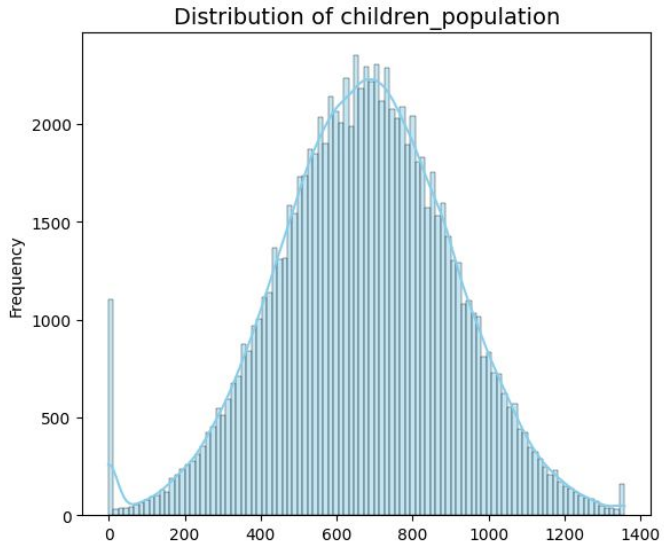
Methodologies + Evolution

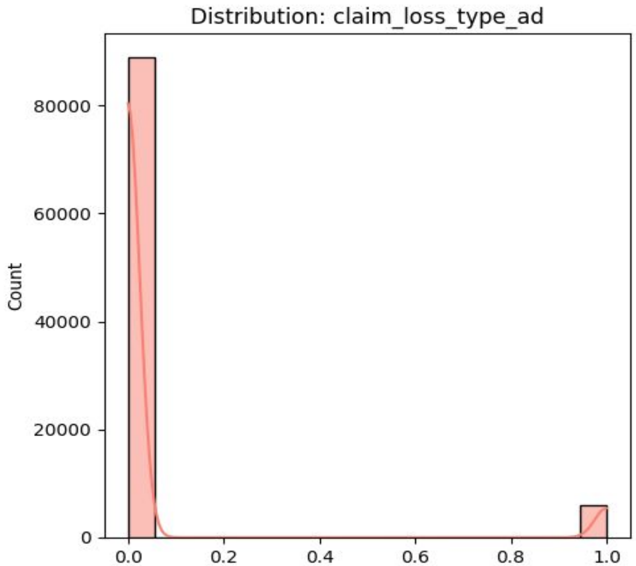
- Implemented 4 strategies for uncertain points selection and compared their results across accuracy and financial cost based metric.
- Formulated a hybrid strategy encompassing their combination.



Data Generation

- Numerical distribution: Gaussian distribution over the [min, max] range, with random noise addition
- Categorical distribution: One-hot labeling.





Weighted cost

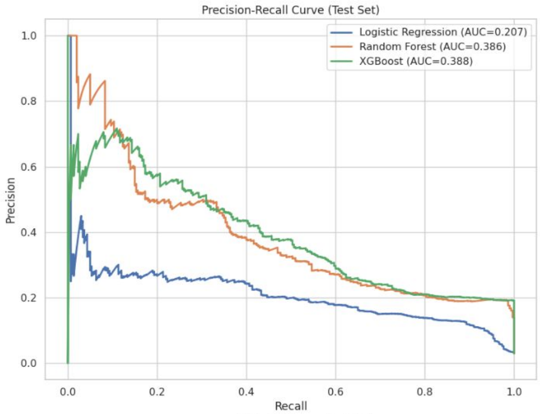
- **COST_TP = 100** ⇔ model says fraud (1), actually fraud (1) ⇒ investigation cost
- **COST_FP = 150** ⇔ model says fraud (1), actually not fraud (0) ⇒ false alarm cost + reputation damage
- **COST_TN = 0** ⇔ model says not fraud (0), actually not fraud (0) ⇒ no cost
- **COST_FN = 0.9 x sum_insured** ⇔ model says not fraud (0), actually fraud (1)⇒ missed fraud = pay out 90% of claim

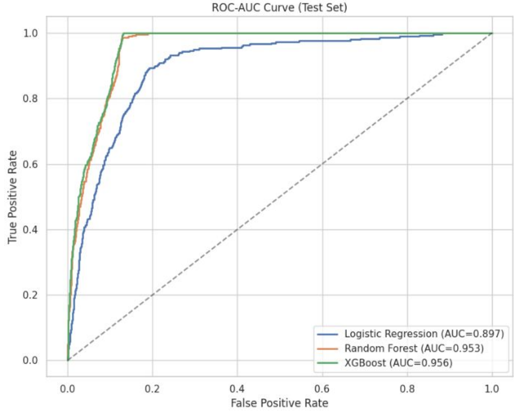
See References B9, B10, B11.

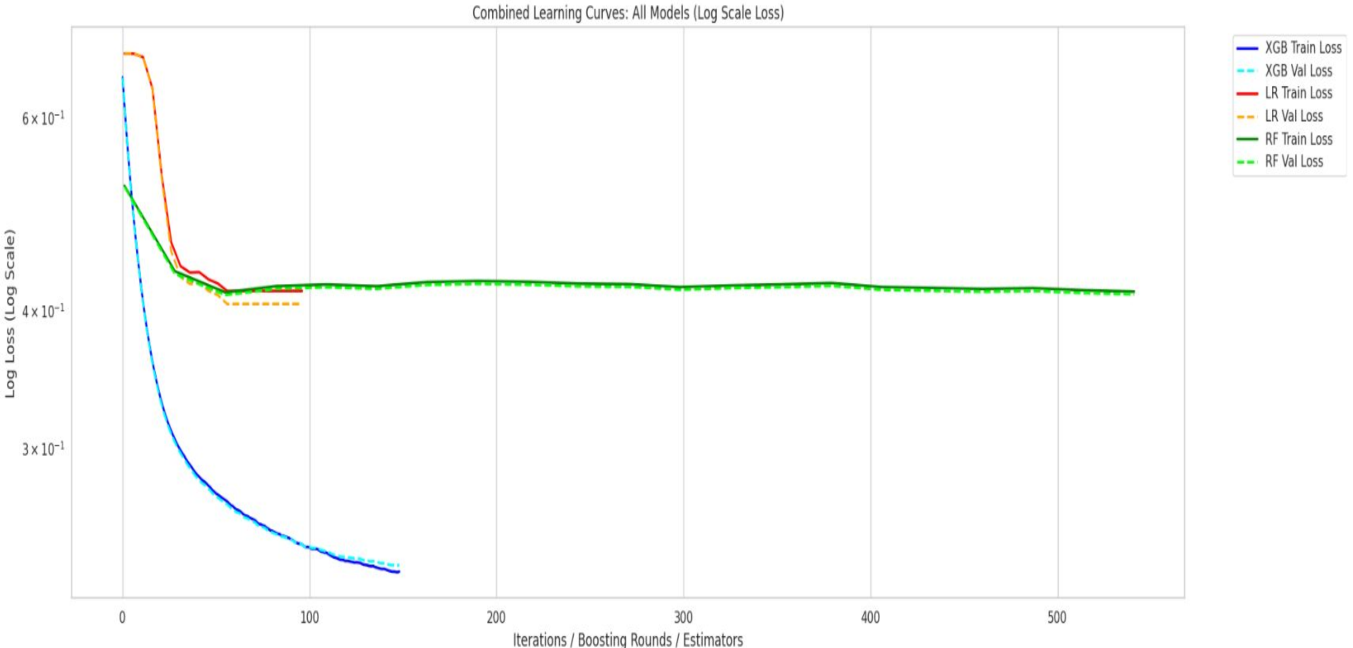
Results - One Shot Training (Accuracy based model)

Train (45k) and val (20k) from labeled, while Test (10k) from unlabeled set

Model	ROC-AUC	F1-Score	Precision	Recall
Logistic Regression	0.897446	0.194435	0.108704	0.920000
Random Forest	0.952515	0.321749	0.193396	0.956667
XGBoost	0.955820	0.321176	0.191932	0.983333





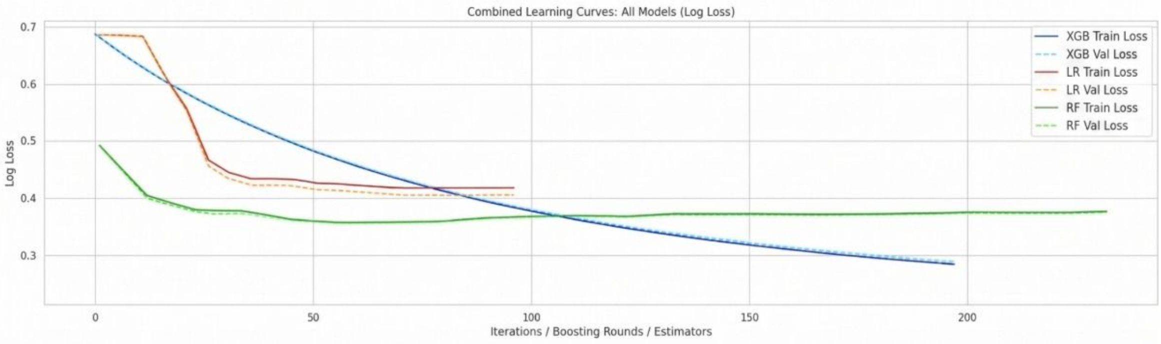


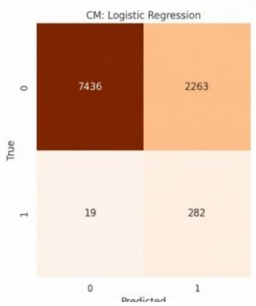
Results - One Shot Training

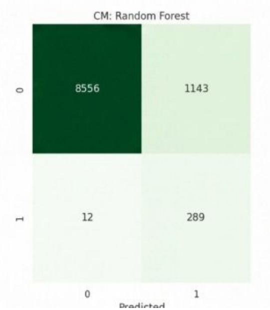
Model	ROC-AUC	F1-Score	Precision	Recall
Logistic Regression	0.897624	0.194435	0.108704	0.920000
Random Forest	0.951370	0.324719	0.195270	0.963333
XGBoost	0.956589	0.321951	0.192233	0.990000

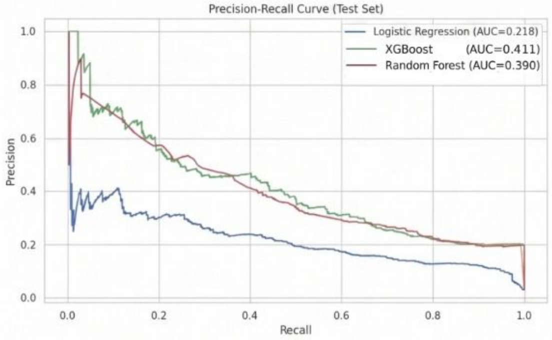
Train (45k) and val (20k) from labeled,
while Test (10k) from unlabeled set

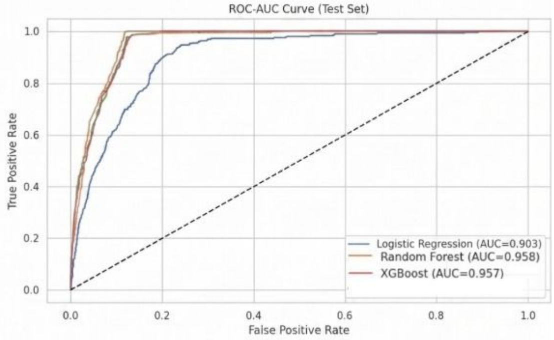
Accuracy based models









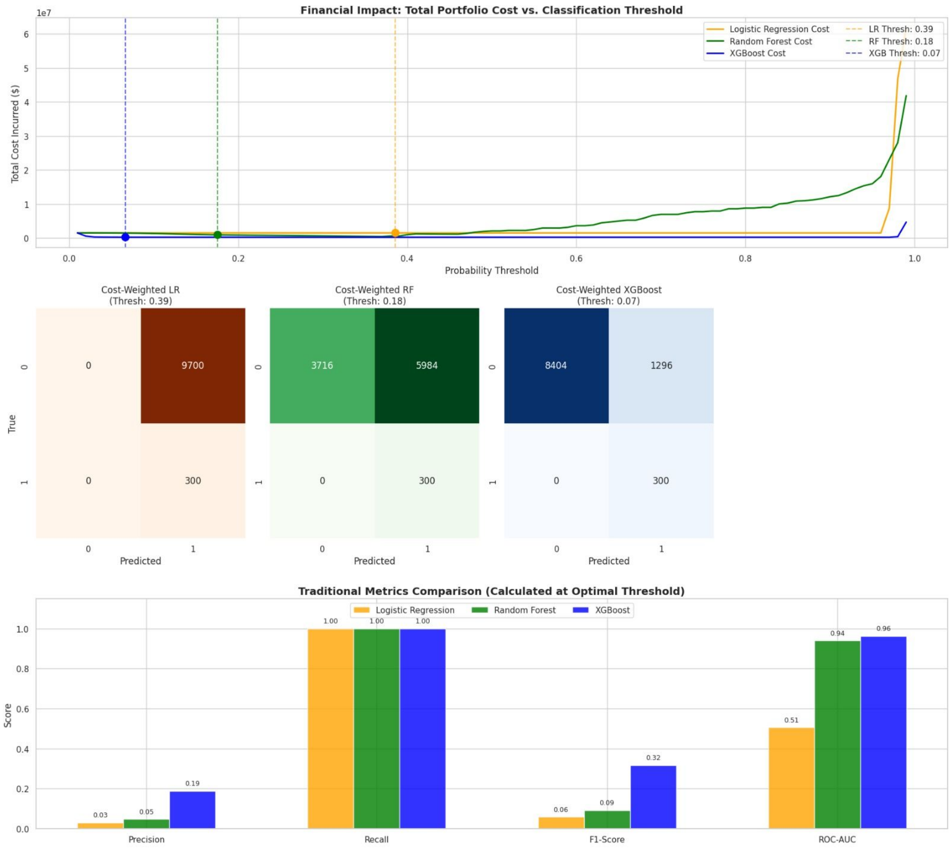


Results - One Shot Training

Train (45k) and val (20k) from labeled, while
Test (10k) from unlabeled set

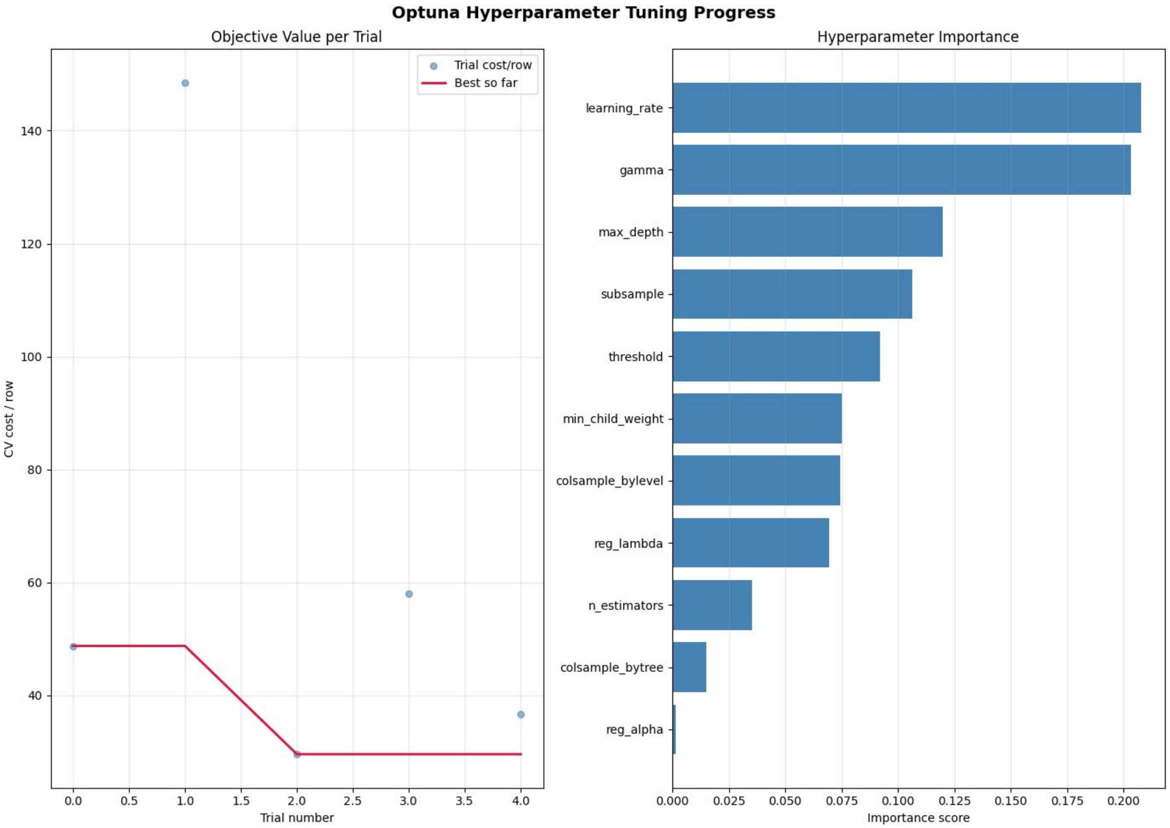
Financial cost based models

	Model	Opt_Threshold	Cost / Row	Total Cost (Test)	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.386005	148.50	1485000.0	0.03000	1.0	0.058252	0.506235
1	Random Forest	0.175799	92.76	927600.0	0.04774	1.0	0.091130	0.940153
2	XGBoost	0.066717	22.44	224400.0	0.18797	1.0	0.316456	0.962196



Grid Search (baseline) model

Case 1: Train (45k), val (10k) and test (10k) from labeled set only

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── Final Test Set Results ──────────────────
  Threshold              : 0.1085
  Best boost round       : 237
  Total business cost    : 225,250.00
  Cost per row           : 22.5250
  TP / FP / TN / FN     : 301 / 1301 / 8398 / 0
  Precision              : 0.1879
  Recall                 : 1.0000
  F1                     : 0.3163
────────────────────────────────────────────
```

```
XGB params:
  max_depth                  = 5
  min_child_weight           = 4
  gamma                      = 0.45606998421703593
  learning_rate              = 0.12448918446337819
  subsample                  = 0.5998368910791798
  colsample_bytree           = 0.708540663048167
  colsample_bylevel          = 0.7554487413172255
  reg_alpha                  = 2.429880728901692e-08
  reg_lambda                 = 0.0019275890163896973
```

Various Active Learning strategies

- **Cost-Weighted:** Selects highly uncertain predictions and scales them by their financial impact (insured amount) to prioritize reviewing costly borderline cases.
- **Uncertainty:** Selects the most ambiguous samples closest to the 50% decision boundary, where the model is least confident.
- **Diversity (Cluster + Uncertainty):** Selects a broad range of hard-to-predict samples by combining the model's uncertainty with spatial distances from K-means clusters to avoid querying redundant data points.
- **Expected Cost Reduction:** Selects samples with the highest potential financial regret by weighing the specific monetary penalties of making a false positive versus a false negative.

AL (various strategies) vs Hybrid AL comparison on Out-of-Sample (OOS) test

Head-to-head comparison on final held-out test set

model	AL (Simple)	AL (Cost-Weighted)	AL (Diversity)	Hybrid AL
auc	0.9563	0.9558	0.9572	0.9570
biz_cost	**589,200.00**	**589,350.00**	**587,100.00**	**587,250.00**
precision	0.1996	0.1996	0.2003	0.2002
recall	1.0000	1.0000	1.0000	1.0000
f1	0.3328	0.3327	0.3337	0.3337
train_size	46,000	46,000	46,000	**45,675**

Δ Hybrid AL vs AL (Cost-Weighted)
AUC : +0.0012
Business cost: -2,100.00 ✅ cheaper
Recall : +0.0000
F1 : +0.0010

Δ Hybrid AL vs AL (Simple)
AUC : +0.0007
Business cost: -1,950.00 ✅ cheaper
Recall : +0.0000
F1 : +0.0009

Δ Hybrid AL vs AL (Diversity)
AUC : -0.0002
Business cost: +150.00 ❌ more expensive
Recall : +0.0000
F1 : +0.0000

```
df_unlabeled total rows      : 30,000
seen rows (union, deduped)   : 2,417
final held-out test rows     : 27,583
```

AL (various strategies) vs Hybrid AL comparison on Out-of-Sample (OOS) test

(remove expected cost strategy (it show performance instead of improvement))

Head-to-head comparison on final held-out test set (27,583 rows)

model	AL (Simple)	AL (Cost-Weighted)	AL (Diversity)	AL (Expected Cost)	Hybrid AL
auc	0.9563	0.9558	0.9572	0.9567	0.9570
biz_cost	**589,200.00**	**589,350.00**	**587,100.00**	**588,600.00**	**587,250.00**
precision	0.1996	0.1996	0.2003	0.1998	0.2002
recall	1.0000	1.0000	1.0000	1.0000	1.0000
f1	0.3328	0.3327	0.3337	0.3331	0.3337
train_size	46,000	46,000	46,000	46,000	**45,675**

Δ Hybrid AL vs AL (Expected Cost)
AUC : +0.0003
Business cost: -1,350.00 ✅ cheaper
Recall : +0.0000
F1 : +0.0006

Δ Hybrid AL vs AL (Cost-Weighted)
AUC : +0.0012
Business cost: -2,100.00 ✅ cheaper
Recall : +0.0000
F1 : +0.0010

Δ Hybrid AL vs AL (Simple)
AUC : +0.0007
Business cost: -1,950.00 ✅ cheaper
Recall : +0.0000
F1 : +0.0009

Δ Hybrid AL vs AL (Diversity)
AUC : -0.0002
Business cost: +150.00 ❌ more expensive
Recall : +0.0000
F1 : +0.0000

```
df_unlabeled total rows      : 30,000
seen rows (union, deduped)   : 2,417
final held-out test rows     : 27,583
```

Baseline vs AL vs NSL vs Hybrid AL comparison on Out-of-Sample (OOS) test

Head-to-head comparison on final held-out test set (27,707 rows)

model	Baseline (labeled pool only)	Non-Sequential (random 1k unlabeled)	Active Learning (Cost-weighted Uncertainty)	Hybrid AL (Multi-Strategy)
auc	0.9553	0.9561	0.9571	0.9563
biz_cost	617,400.00	616,800.00	608,250.00	607,950.00
precision	0.1917	0.1919	0.1944	0.1945
recall	1.0000	1.0000	1.0000	1.0000
f1	0.3218	0.3220	0.3255	0.3257

Δ Hybrid AL vs AL (Expected Cost)
AUC : +0.0003
Business cost : -1,350.00 ✅ cheaper
Recall : +0.0000
F1 : +0.0006

df_unlabeled total rows : 30,000
seen rows (union, deduped) : 2,293
final held-out test rows : 27,707

References

A] UK Data Sources Used

1. **Population, Occupation, Housing:**
   - **1.1.** ONS Labour Force Survey
   - 1.2. UK Housing Survey 2023
   - 1.3. Gov.uk English Housing Data
2. **Risk Maps:**
   - **2.1.** Environment Agency Flood Risk Map
   - 2.2. Met Office Severe Weather Patterns
3. **Insurance Market:**
   - **3.1.** Association of British Insurers (ABI) Home Insurance Report
   - 3.2. Insurance Fraud Bureau (IFB) Fraud Typology
   - 3.3. FCA Household Insurance Market Analytics
4. **Payment Behaviour:**
   - **4.1.** Experian UK Financial Wellness Index
   - 4.2. FCA Consumer Risk Survey

B] Academic & Industry Sources

1. **Mordor Intelligence** (2024). *UK Home Insurance Market Report*. https://www.mordorintelligence.com/industry-reports/uk-home-insurance-market
   - 1.1. Property type distribution, premium ranges
2. **Mintel** (2024). *UK Home Insurance Market Report 2024*. https://store.mintel.com/report/uk-home-insurance-market-report-2024
   - 2.1. Market size (£9.4B forecast), ownership rates
3. **GlobalData** (2024). *UK Household Insurance Market Dynamics*.
   - 3.1. GWP growth, premium inflation, claims landscape
4. **Confused.com** (2024). *UK Home Insurance Statistics*. https://www.confused.com/home-insurance/home-insurance-statistics
   - 4.1. Average premiums (£396), claim frequencies
5. **Uswitch** (2025). *Home Insurance Statistics*. https://www.uswitch.com/home-insurance/studies/home-insurance-statistics-uk/
   - 5.1. Weather claims (£585M), regional variations
6. **Association of British Insurers (ABI)** (2024). *Insurance Fraud Statistics*.
   - 6.1. 10-15% fraud rate benchmark
7. **Office for National Statistics (ONS)** (2021). *UK Census - Housing Stock*.
   - 7.1. Property age distribution, urban/rural split
8. **English Housing Survey** (2023). *Housing Stock Report*.
   - 8.1. Property condition, age bands
9. Economic and social cost of fraud 2023 to 2024
10. Uncovering the real cost of fraud
11. Insurance Fraud Taskforce: interim report

C] Regulatory & Government Sources

1. **Financial Conduct Authority (FCA)** (2024). *Home Insurance Pricing Practices*.
   - 1.1. Auto-renewal regulations, price walking reforms
2. **Environment Agency** (2024). *Flood Risk Assessment*.
   - 2.1. 10% of English homes in flood zones

THANK YOU

Next Steps

- Analyze performance of neural networks in comparison to AL.
- Weighted average of Financial cost and accuracy for a new loss function taking into account both the aspects.

- **Cost-Weighted:** Selects highly uncertain predictions and scales them by their financial impact (insured amount) to prioritize reviewing costly borderline cases.
- **Uncertainty:** Selects the most ambiguous samples closest to the 50% decision boundary, where the model is least confident.
- **Diversity (Cluster + Uncertainty):** Selects a broad range of hard-to-predict samples by combining the model's uncertainty with spatial distances from K-means clusters to avoid querying redundant data points.
- **Expected Cost Reduction:** Selects samples with the highest potential financial regret by weighing the specific monetary penalties of making a false positive versus a false negative.